

Luis Angel Guerrero Hoyos

Professional Summary

GIS Specialist with 6+ years of experience delivering cartographic products, geospatial analysis, and spatial data management for dams, energy, and natural-resources engineering projects. Proficient in ArcGIS Pro, ArcGIS Online, and Esri web applications, with hands-on experience processing elevation data (LiDAR point clouds, DEMs), Sentinel satellite imagery, and vector/raster geospatial formats (geodatabases, GeoJSON, GeoPackage, Shapefile). Skilled in Python and SQL for geospatial automation and GNSS-based field data collection. Holds an MSc in Civil Engineering (University of Washington) and a BSc in Geological Engineering.

Skills

- **Esri ArcGIS Software Stack:** ArcGIS Pro (cartography, geoprocessing, Spatial Analyst), ArcGIS Online, Experience Builder, Dashboards
- **Geospatial Data & Formats:** Geodatabases, GeoJSON, GeoPackage, Shapefile, GeoTIFF rasters, DEMs, point clouds, Sentinel satellite imagery, spatial projections & CRS
- **Remote Sensing & Elevation Processing:** LiDAR (ICP registration, M3C2), InSAR/SAR processing (ISCE+, Dolphin, PyGMTSAR), QGIS, CloudCompare, PDAL
- **Programming & Scripting:** Python (GeoPandas, Rasterio, GDAL), Arcade, SQL, GNSS/GPS field data collection
- **Databases:** Geodatabases, SQL, PowerBI
- **Soft Skills:** Cross-disciplinary collaboration, technical communication, organization and data management, adaptability

Experience

- Staff Engineer – GIS & Remote Sensing (July 2025 – Present)*

Shannon and Wilson Inc. 

 - Develop professional cartographic products and maps in ArcGIS Pro for engineering programs, managing enterprise geodatabases, layer files, and metadata for an oil-industry pipeline monitoring program and a hydropower dam-monitoring program.
 - Lead development of SWLC3D, a LiDAR point-cloud change-detection workflow (ICP registration and M3C2 differencing) using PDAL, CloudCompare (cloudcompy), and arcpy, producing displacement-vector shapefiles and raster deliverables.
- Research Assistant & Geospatial Scientist (August 2023 – June 2025)*

University of Washington 

 - Investigated geospatial relationships between earthquake-induced landslides and reactivations, integrating elevation data (LiDAR DEMs, point clouds), Sentinel-1 satellite imagery, and vector datasets (GeoJSON, GeoPackage, Shapefile) at regional scale.
 - Performed spatial analysis and raster/vector processing using Python (GeoPandas, Rasterio, GDAL) and produced cartographic products for research and agency stakeholders.
 - Completed InSAR Processing and Analysis (ISCE+) training through EarthScope Consortium & NSF (2024).
- GIS & InSAR Monitoring Analyst (July 2021 – August 2023)*

Hexagon Geospatial 

 - Delivered real-time geospatial monitoring of open-pit mine slopes and tailings dams, integrating GNSS field data collection with ground-based radar into centralized geodatabases and maintaining data quality across multi-sensor sources.
 - Produced daily, weekly, and monthly cartographic reports supporting client decision-making on slope and dam stability across mining operations in Latin and North America.
- Geological Engineer – GIS & Resource Data Management (January 2019 – July 2021)*

South32 

 - Managed enterprise GIS geodatabases, including metadata, for mineral resource estimation; coordinated field data collection for drilling campaigns and produced cartographic deliverables for decision-making.

Education

- Master of Science in Civil Engineering
University of Washington
June 2025
- Bachelor of Science in Geological Engineering
Universidad Nacional de Colombia (UNALMED)
May 2019

Awards & Certifications

- InSAR Processing and Analysis (ISCE+) – EarthScope Consortium & NSF, August 2024
- CEE Departmental Graduate Fellowship – University of Washington, September 2023
- InSAR and RAR Radar Monitoring – University of Arizona, December 2020

Languages

- English
- Spanish (Native)